

SWE 307 BIG DATA PROJECT - 2

Redis for RDB caching, HDFS as Data Storage and a Local LLM model for chatting

Due date: 22.10.2026 Thursday, in class.

This project has multiple objectives:

- 1) Build a real-time information system that combines the speed of in-memory data processing with the reliability and scalability of distributed storage (See Figure 1). The system uses **Redis** to handle fast, real-time ingestion and temporary caching of incoming data.
- 2) Use of a distributed file server practically unlimited size in an information system as a document server (or object storage like S3 in AWS). **Hadoop-HDFS** is an open source distributed, fault-tolerant file system that is used for storing documents in information systems. The Hadoop HDFS is expected to improve response time as well.
- 3) Experimenting a local LLM server. Ollama with HTTP server is added to the system just for chatting from front-end.

Key Objectives:

- Use scott database (<https://github.com/rsp/pg-scott>) for RDBMS(MySQL). At the beginning, import data from given site to your database.
- Use **Redis** to temporarily store and process incoming data for fast access. Periodically flush the data from Redis to **RDBMS** for durable storage and future processing. Show that the caching mechanism works as expected. (See example: <https://www.youtube.com/watch?v=0a-RIJx09rg>)
- Use **Hadoop-HDFS** for document storage. Please see examples done in the class.
- Install **Ollama server and llama 3.2 LLM model** as local LLM server. One LLM model (llama3.2:latest) would be sufficient. This server can be tested via CLI but it should communicate with your Java web application.

PS: Installing Ollama server and llama3.2 model takes some time depending on network speed and takes a lot of disk space.

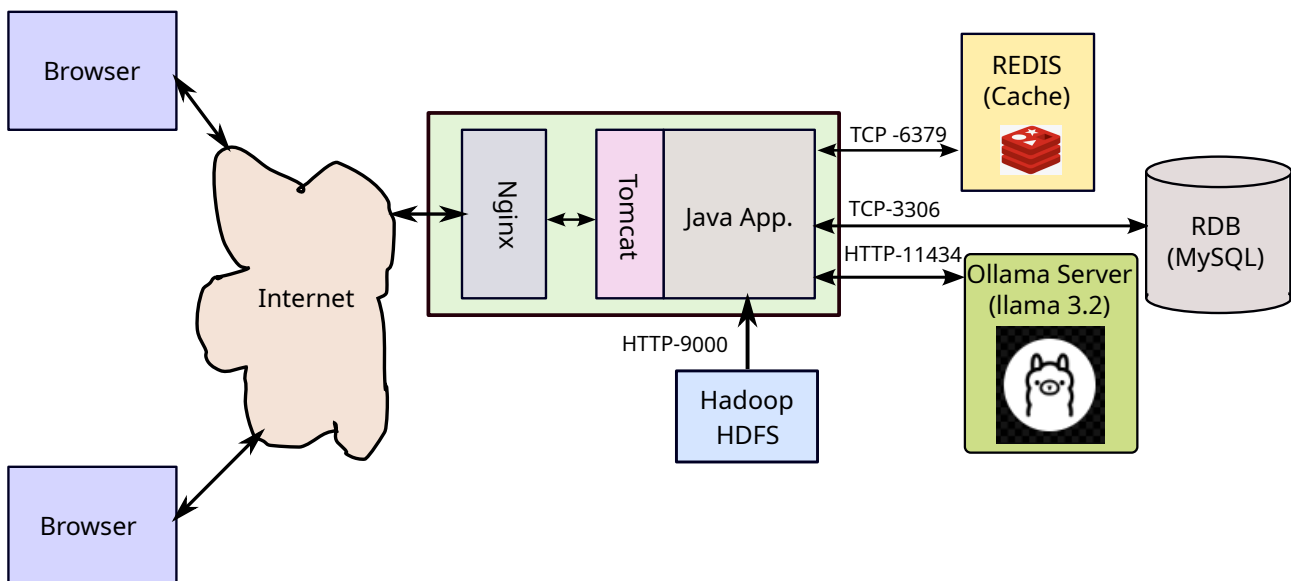


Figure 1. Architectural block diagram of project 2.

What is required from you is as follows:

- 1) Single node Hadoop-HDFS cluster must be installed in your computer.

- 2) Redis and MySQL must be installed on your computer.
- 3) Ollama server with llama 3.2 model must be installed on your computer.
- 4) A simple Java Spring-Boot application will be developed to perform the following tasks:
 - a) Personnel and department data will be read/updated from the database via Redis.
 - b) Personnel images will be stored to/read from HDFS.
 - c) There will be two pages in this application: 1) Personnel information will be displayed in a table using the JOIN operation on database. Information to display: employee image, employee name, manager name, salary, commission, department name.
 - d) On the second page (chat page), we should be able to chat with LLM server. Show us some chat outcome comes from local LLM model runs under Ollama server.

Notes:

- 1) Some groups may want to implement the project using G-Drive or AWS-S3 Object storage instead of HDFS, these implementations will also be accepted. This will be discussed in the class.
- 2) Example image files and data files will be provided on Github repository, you can clone/download everything provided.

Link: <https://github.com/ozmen54/SWE307-2026.git>

Details will be discussed in the class!